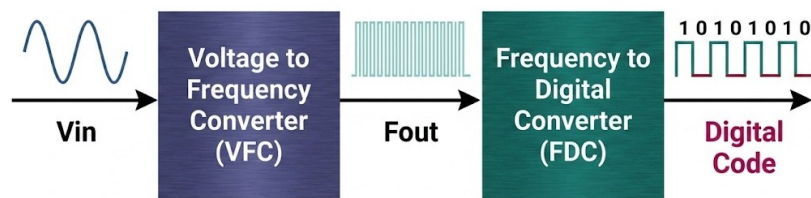


Inside a Current-Starved VCO-Based ADC

A Design Walkthrough from First Principles to 90nm CMOS

Voltage → Frequency → Digital Code

Time-domain analog-to-digital conversion using a five-stage current-starved ring VCO and a four-bit asynchronous-reset counter.



System-level architecture of the VCO-based ADC

Technology	90nm CMOS	Resolution	4-bit
Architecture	VCO-based ADC	Conversion	Time-domain

1 Why Does This ADC Matter?

Imagine wearing an ECG patch for several days.

It continuously watches the tiny electrical signals produced by your heart. You expect it to be small, comfortable, and able to run for a long time without constantly charging it.

Now consider something even more demanding: an implanted medical device. Once it is inside the body, replacing its battery is no longer a simple inconvenience. Every part of the circuit has to justify the energy it consumes.

This is where a surprisingly small circuit becomes important: the **analog-to-digital converter (ADC)**.

Sensors see the physical world as analog signals — voltages and currents that vary continuously. Computers do not. A processor wants numbers.

The ADC is the bridge between these two worlds:

Physical signal $\longrightarrow$ **ADC** $\longrightarrow$ **Digital data**

That bridge has to operate every time the system takes a measurement. So even if the ADC is only one block inside a much larger medical device, its power consumption, area, speed, and accuracy can directly affect the overall system.

1.1 Why Not Just Use a Conventional ADC?

There are already excellent ADC architectures.

A flash ADC is extremely fast, but its comparator count grows as

$$2^N - 1$$

for an N -bit converter.

A SAR ADC is much more area-efficient, but it requires a precision DAC and comparator-based decision process.

A sigma-delta ADC can achieve excellent resolution, but it relies on oversampling and digital filtering, which can become expensive when power and bandwidth are tightly constrained.

For a large piece of laboratory equipment, these trade-offs may be acceptable.

For a tiny battery-powered sensor that is expected to work continuously, they become much more interesting.

1.2 The Question Behind This Project

Instead of asking,

“How can we measure this voltage more precisely?”

we ask a different question:

What if we don't measure the voltage directly at all?

What if we first turn the voltage into something that digital logic is naturally good at measuring — **time**?

That question leads to the architecture explored in this project.

2 The Trick: Turn Voltage Into Time

The idea is surprisingly simple.

Instead of asking the ADC to determine the exact value of V_{in} directly, we make the voltage control the speed of an oscillator.

A higher input voltage makes the oscillator run faster. A lower input voltage makes it run slower.

Now the voltage has become a frequency:

$$V_{in} \longrightarrow f_{VCO}. \quad (1)$$

Once the information is represented as frequency, the rest of the problem becomes digital.

We simply count how many oscillator edges arrive during a known sampling interval:

$$f_{VCO} \longrightarrow N_{\text{count}}. \quad (2)$$

The complete conversion is therefore

Voltage $\rightarrow$ Frequency $\rightarrow$ Digital Code

This is the central idea behind the VCO-based ADC developed in this project.

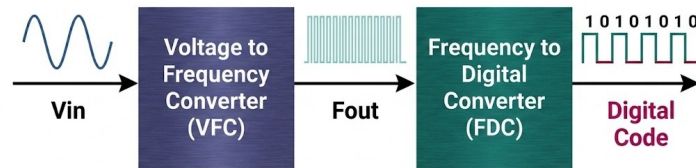


Figure 1: Conceptual flow of the voltage-to-frequency-to-digital conversion process.

The important part is what this changes architecturally.

The analog section no longer has to perform the entire analog-to-digital conversion by itself. It only has to create a measurable timing signal.

The digital section then does what digital CMOS is naturally good at: **counting, storing, and processing transitions**.

3 Building the Voltage-to-Frequency Converter

We now have one requirement:

The input voltage must control the speed of an oscillator.

A ring oscillator is a natural way to achieve this.

3.1 Starting With a Ring Oscillator

A ring oscillator is made by connecting an odd number of inverting stages in a loop.

Each inverter tries to change the state of the next one. Because the loop contains an odd number of inversions, the circuit cannot settle to a stable logic state. Instead, the transition continuously travels around the loop.

The time required for the transition to travel through all stages determines the oscillation frequency.

For this project, we use **five stages**.

The important observation is:

More delay per stage $\Rightarrow$ Slower oscillator

and

Less delay per stage $\Rightarrow$ Faster oscillator.

So the next question is obvious:

How do we control the delay using V_{in} ?

3.2 Controlling the Current

Instead of allowing each inverter to switch with its maximum available current, we deliberately limit the current flowing through it.

This is the idea behind a **current-starved ring oscillator**.

The control voltage determines the available bias current. That current controls how quickly the capacitances at the inverter nodes can charge and discharge.

Therefore,

$$V_{ctrl} \rightarrow I_{bias} \rightarrow \text{switching delay} \rightarrow f_{VCO}. \quad (3)$$

This is the mechanism that turns a voltage into frequency.

An approximate relationship for the oscillator frequency is

$$f_{osc} = \frac{I_{bias}}{2NC_{total}V_{DD}} \quad (4)$$

where $N = 5$ is the number of ring-oscillator stages and C_{total} represents the effective load capacitance associated with each stage.

Over the useful tuning range, the VCO characteristic can be approximated as

$$f_{out}(V_{in}) = f_0 + K_{VCO}V_{in} \quad (5)$$

where K_{VCO} represents the VCO gain in MHz/V and f_0 is the free-running frequency at $V_{in} = 0$.

The idealized equations explain the principle. The actual tuning curve must ultimately be obtained from transistor-level simulation.

3.3 Building the Five-Stage VCO

A practical reconstruction sequence is:

1. Create the five inverter stages.
2. Connect the stages in a ring configuration.
3. Introduce the current-starving devices into each stage.
4. Connect the bias/control circuitry to establish the desired current range.
5. Apply the supply voltage and control voltage.
6. Perform transient simulation.
7. Verify that sustained oscillation occurs.
8. Measure the oscillation frequency.
9. Sweep V_{ctrl} or V_{in} across the intended operating range.
10. Extract the resulting tuning characteristic.

4 Now We Have Frequency. How Do We Count It?

The first half of the problem is solved.

The input voltage has become a frequency.

But a frequency is still an analog quantity. The processor eventually needs a number.

Fortunately, digital logic already has a very simple way of measuring frequency:

Count the edges.

If the oscillator produces more edges during a fixed amount of time, the input voltage must have produced a higher frequency.

So the next part of the ADC is a digital counter.

5 Building the Frequency-to-Digital Converter

5.1 The Building Block: A NAND-Based D Flip-Flop

The counter is built from cascaded D flip-flops, each constructed from three-input and two-input NAND gates, with an **active-low asynchronous reset** $\overline{RESET}$.

The asynchronous reset is important because the ripple-counter topology needs to be cleared at a defined point independently of the normal clock edge.

Before connecting the counter to the VCO, the flip-flop should be verified independently.

The essential checks are:

- correct data storage;
- correct clock operation;
- correct logic-state transitions; and
- correct asynchronous reset behaviour.

5.2 Chaining Flip-Flops Into a Counter

Four flip-flops are cascaded in a **ripple**, or asynchronous, arrangement.

Only the first flip-flop is clocked directly by the VCO output. Each subsequent stage is clocked from the output of the preceding flip-flop.

This produces successive divide-by-two behaviour:

Output	Division	Role
D_1	$\div 2$	LSB
D_2	$\div 4$	
D_3	$\div 8$	
D_4	$\div 16$	MSB

A NAND gate monitors the running count and asserts $\overline{RESET}$ when the required reset condition is reached, clearing the flip-flops and preparing the counter for the next sampling interval.

5.3 Turning Counts Into a Digital Code

Within a fixed sampling window,

$$T_s = \frac{1}{f_s}, \quad (6)$$

the counter tallies the relevant edges of the VCO output.

The digital output can therefore be represented as

$$N = \lfloor f_{out}(V_{in})T_s \rfloor \quad (7)$$

where N represents the number of counted edges during the sampling window.

This equation captures the central operation of the digital section.

A higher frequency produces more counted edges.

More counted edges produce a larger digital value.

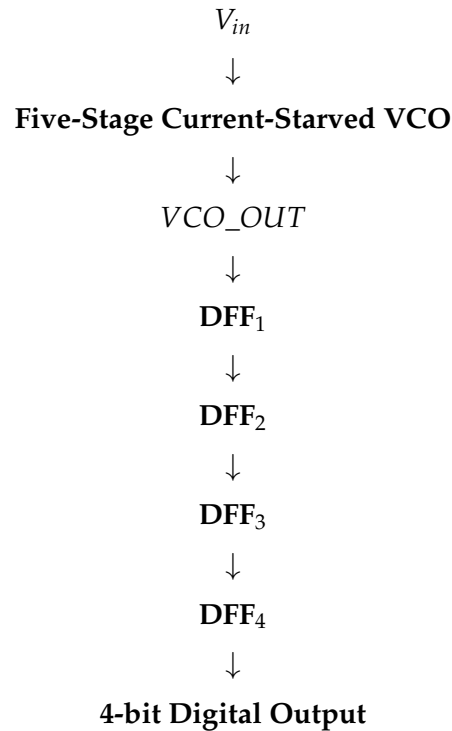
Therefore,

$$V_{in} \uparrow \Rightarrow f_{VCO} \uparrow \Rightarrow N \uparrow. \quad (8)$$

6 Putting the Two Worlds Together

The complete ADC is obtained by connecting the VCO output directly to the clock input of the first counter stage.

The complete signal path is



The reset network controls the start and termination of each counting window.

At this point, the original analog input has travelled through two different representations:

Voltage $\rightarrow$ Frequency $\rightarrow$ Count

That is the complete idea of the converter.

7 Does It Actually Work?

An architecture is only useful if the circuit behaves as expected.

To verify the complete chain, representative input voltages are applied and the resulting oscillator frequency and digital output are observed.

The reported operating points are:

V_{in} (V)	f_{out}	Digital Output
0.4	~ 160 MHz	0001
0.6	~ 230 MHz	0110
0.8	~ 330 MHz	1011
1.0	~ 470 MHz	1101

The important observation is not simply any individual number.

It is the trend:

Higher input voltage $\rightarrow$ higher VCO frequency $\rightarrow$ higher digital code.

The reported checkpoints show a monotonic increase in the digital output with input voltage.

This is the essential functional behaviour expected from the voltage-to-frequency-to-digital conversion chain.

The reset circuitry is also intended to clear the counter at the defined sampling boundary, preparing the system for the next conversion interval.

8 How Was the Circuit Verified?

The project follows a schematic-driven transistor-level simulation flow.

The purpose of verification is not to jump immediately to the complete ADC. Each building block is checked first and then integrated.

8.1 Required Environment

A researcher attempting to reproduce the design requires:

- Cadence Virtuoso or an equivalent IC design environment;
- access to an appropriate 90nm CMOS technology PDK;
- transistor-level simulation capability;
- transient analysis;
- waveform measurement capability; and
- a method for extracting oscillator frequency from the simulated waveform.

The proprietary software and PDK themselves are outside the scope of the repository.

8.2 Verification Sequence

The recommended verification sequence is:

1. Verify a single inverter.
2. Verify the five-stage ring oscillator.
3. Verify sustained oscillation.
4. Measure the VCO frequency.
5. Sweep the control voltage.
6. Generate the VCO tuning characteristic.
7. Verify the D flip-flop.
8. Verify asynchronous reset operation.
9. Cascade the four flip-flops.
10. Verify divide-by-two behaviour.
11. Verify the reset logic.
12. Connect the VCO to the counter.
13. Run the complete transient simulation.
14. Sweep representative input voltages.
15. Compare the measured frequency and digital output with the expected trend.

8.3 Simulation Configuration

The documented project workflow uses transient simulation with a reported run duration of approximately 35 ns and a maximum timestep of approximately 5 ps.

These values describe the configuration used for the documented simulation workflow. They should not be interpreted as universal simulation requirements for every implementation.

9 What Are the Engineering Trade-offs?

The architecture does not make the difficult parts of ADC design disappear.

It moves them.

The VCO-based approach makes the analog section relatively compact and allows digital circuitry to perform the frequency measurement. However, the VCO itself remains sensitive to several non-idealities.

9.1 VCO Linearity

The relationship between V_{in} and f_{out} is not necessarily perfectly linear across the complete operating range.

The useful ADC range should therefore be selected from the measured VCO tuning characteristic rather than relying exclusively on the ideal linear approximation.

9.2 Timing and Counter Speed

The counter must operate reliably at the highest expected VCO frequency.

The flip-flop propagation delay, setup/hold requirements, ripple propagation, and reset timing therefore become increasingly important as the operating frequency increases.

9.3 Sampling Window

The sampling period directly determines how many VCO edges can be counted.

From

$$N = \lfloor f_{out} T_s \rfloor ,$$

a longer sampling window provides more opportunities for counting edges, but it also reduces the maximum conversion rate.

This creates a fundamental trade-off between:

resolution and **conversion speed.**

9.4 Process, Voltage, and Temperature

The VCO is an analog timing element.

Its frequency can change with:

- transistor process variation;
- supply voltage;
- temperature;
- device mismatch;

- capacitive loading; and
- layout parasitics.

Therefore, technology scaling does not automatically guarantee a better ADC.

The digital portion may benefit significantly from scaling, but the VCO still requires careful design and verification.

10 What Happens When CMOS Gets Smaller?

One of the interesting questions behind this project is what happens when the same architecture is moved to a smaller CMOS technology.

The architecture itself does not fundamentally need to change.

However, transistor characteristics, capacitance, supply voltage, switching speed, and power all change with technology scaling.

Using the scaling relationships adopted in this project, with S as the feature-size scaling factor,

$$S = \frac{L_{ref}}{L_{new}}, \quad (9)$$

$$f_{osc,new} \approx S f_{osc,ref}, \quad (10)$$

$$P_{new} \approx \frac{1}{S^2} P_{ref}. \quad (11)$$

For the reported migration from the 130nm reference design to 90nm,

$$S = \frac{130}{90} \approx 1.44. \quad (12)$$

The comparison reported for the project is:

Parameter	130nm (Reference)	90nm (This Work)	45nm (Projection)
Supply V_{DD}	1.2 V	1.2 V	1.0 V
VCO f_{min}	100 MHz	160 MHz	130 MHz
VCO f_{max}	1.6 GHz	470 MHz	350 MHz
Sampling rate f_s	100 MHz	60 MHz	50 MHz
Resolution	4-bit	4-bit	4-bit
Power	0.257 mW	0.214 mW	0.198 mW
Active area	0.007 mm ²	≈0.0033 mm ²	≈0.0018 mm ²
SNDR	62 dB	65 dB	68 dB
Walden FoM	125 fJ/step	105 fJ/step	62 fJ/step

Important: the 130nm column represents the reference design, the 90nm column represents the work documented here, and the 45nm column is a projection rather than a fabricated or experimentally verified 45nm implementation.

The purpose of this comparison is therefore to illustrate the expected technology-scaling trend rather than to claim that all three designs were physically implemented.

11 Where the Current Design Stops

The current work establishes the architecture and schematic-level transient behaviour.

But a circuit that works in a schematic simulation is not yet a finished chip.

In the physical implementation, wires introduce resistance and capacitance. Device mismatch changes transistor behaviour. Temperature changes device characteristics. Process variation shifts operating points.

For that reason, several important steps remain.

1. **Technology-specific transistor resizing**

The transistor dimensions should be optimized specifically for the target 90nm technology rather than relying entirely on ratios inherited from the reference design.

2. **Post-layout parasitic extraction**

Layout parasitics must be extracted and included in simulation because schematic-level simulations can underestimate delay and power.

3. **Full end-to-end verification**

The complete ADC should be tested across a substantially larger input range rather than relying only on a few representative checkpoints.

4. **PVT analysis**

Process, voltage, and temperature corners should be evaluated to determine whether the converter maintains its intended behaviour across environmental and manufacturing variation.

5. **Monte Carlo analysis**

Statistical simulations can be used to study the impact of device mismatch and process variation.

6. **Physical verification**

A complete implementation should include layout, design-rule checking, layout-versus-schematic verification, and extracted post-layout simulation.

12 How to Reproduce the Project

The purpose of this repository is not to distribute a proprietary Cadence environment.

Instead, it provides the information required to understand and reconstruct the design in an appropriate IC design environment.

12.1 Step 1 — Prepare the Technology Environment

Obtain access to an appropriate 90nm CMOS PDK and create a design library associated with that technology.

The exact PDK installation procedure is technology-provider specific and is therefore not reproduced here.

12.2 Step 2 — Build the VCO

Construct the five-stage current-starved ring oscillator.

Verify that it oscillates correctly before proceeding to the digital section.

12.3 Step 3 — Characterize the VCO

Apply the control voltage and perform transient simulations.

Measure the oscillation frequency.

Repeat the measurement over the intended input range and generate the VCO tuning characteristic.

12.4 Step 4 — Build and Verify the D Flip-Flop

Construct the NAND-based D flip-flop.

Verify data transfer, clock operation, and active-low asynchronous reset.

12.5 Step 5 — Build the Four-Bit Counter

Cascade the four flip-flops.

Verify:

$$f_{D_1} = \frac{f_{VCO}}{2}, \quad f_{D_2} = \frac{f_{VCO}}{4},$$

$$f_{D_3} = \frac{f_{VCO}}{8}, \quad f_{D_4} = \frac{f_{VCO}}{16}.$$

12.6 Step 6 — Add Reset Logic

Connect the NAND-based reset logic to the appropriate counter outputs.

Verify that the counter returns to its initial state when the reset condition is asserted.

12.7 Step 7 — Integrate the ADC

Connect

$$VCO_OUT \rightarrow CLK_{DFF1}. \quad (13)$$

Connect the remaining flip-flops in the ripple configuration and connect the reset network.

12.8 Step 8 — Perform End-to-End Simulation

Apply representative input voltages.

Measure:

- VCO output frequency;
- counter output transitions;
- reset operation; and
- final digital output code.

The resulting measurements should demonstrate the expected voltage-to-frequency-to-code relationship.

13 Repository Structure

A clean repository can be organized as follows:

```
vco-based-adc-90nm/
|
|-- README.md
|-- LICENSE
|
|-- paper/
|   '-- project_paper.pdf
|
|-- images/
|   |-- vffc.jpeg
|   |-- vco_schematic.png
|   |-- vco_waveform.png
|   |-- tuning_curve.png
|   |-- dff_schematic.png
|   |-- counter_schematic.png
|   '-- adc_waveform.png
|
'-- references/
    '-- references.md
```

The repository should contain original documentation, explanatory material, exported figures, and other files that can legally be redistributed.

The following should remain outside the repository:

- Cadence installation files;
- foundry PDKs;
- proprietary model cards;
- technology files;
- licensed standard-cell or device libraries;
- license-server credentials;
- proprietary simulation databases; and
- any other material restricted by software, foundry, or institutional licensing.

14 The Bigger Picture

The interesting part of this project is not simply that a ring oscillator can be used to build an ADC. The deeper idea is architectural. An ADC normally looks like an analog problem:

Measure a voltage precisely.

This project approaches it as a timing problem:

Make the voltage control time, then measure the time digitally.

That change in viewpoint is what makes the architecture interesting.

The analog circuitry creates a timing signal.

The digital circuitry performs the measurement.

As CMOS technology continues to scale, this boundary between analog and digital becomes increasingly useful to explore.

At the same time, the project exposes the engineering challenges that come with the idea: VCO nonlinearity, jitter, PVT sensitivity, counter timing, sampling-window limitations, and layout parasitics. So the goal is not to claim that a VCO-based ADC is universally better than every conventional ADC. The goal is to understand a different way of building the analog-to-digital boundary and to investigate how far that idea can be pushed in scaled CMOS.

15 Reference

D. M. Ellaithy, "Voltage-controlled oscillator based analog-to-digital converter in 130-nm CMOS for biomedical applications," *Journal of Electrical Systems and Information Technology*, vol. 10, no. 38, Aug. 2023.

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Repository note — This document is a technical guide for the project repository and is separate from the manuscript submitted to the ISED 2026 Conference. It is intended to explain the project architecture, design methodology, simulation workflow, and reproducibility considerations.

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